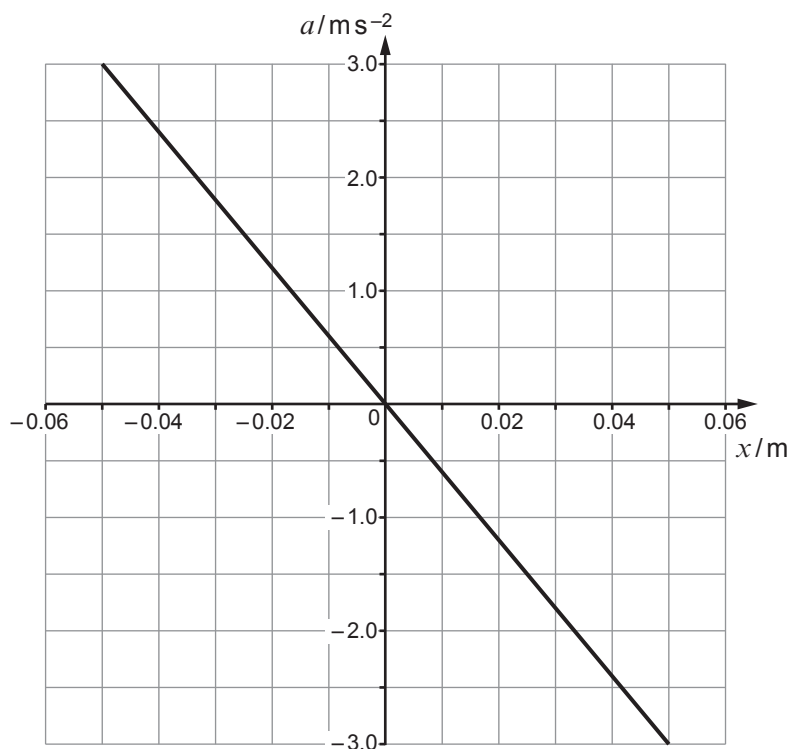


3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a , depends on its displacement, x , from a fixed point.



- (a) (i) Identify the **two** characteristics **of the graph** that show that the object is oscillating with *simple harmonic motion*. [2]

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- (ii) Use the graph to show that the angular frequency, ω , of the oscillation is approximately 7.7 rad s^{-1} . [3]

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(iii) Calculate:

I. the period of oscillation;

[1]

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II. the spring constant;

[2]

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III. the velocity of the object as it passes through the centre of oscillation.

[2]

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(iv) The object was released at the displacement of -0.05 m at time $t = 0$. Sketch on the axes below:

- I. the potential energy of the system (label the sketch PE),
- II. the kinetic energy of the system (label the sketch KE).

Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.)

[3]

Energy



- (b) Explain what is meant by the terms *damping* and *resonance* **and** discuss their importance in real systems giving an example for each.

[6 QER]

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